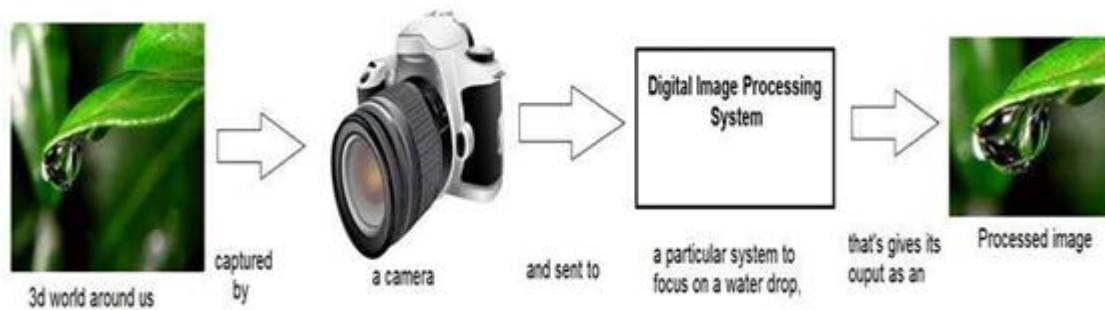


DIGITAL IMAGE FUNDAMENTALS:

The field of digital image processing refers to processing digital images by means of digital computer. Digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels and pixels. Pixel is the term used most widely to denote the elements of digital image.

An image is a two-dimensional function that represents a measure of some characteristic such as brightness or color of a viewed scene. An image is a projection of a 3-D scene into a 2D projection plane.

An image may be defined as a two-dimensional function $f(x,y)$, where x and y are spatial (plane) coordinates, and the amplitude of f at any pair of coordinates (x,y) is called the intensity of the image at that point.



The term **gray level** is used often to refer to the intensity of monochrome images. Color images are formed by a combination of individual 2-D images.

For example: The RGB color system, a color image consists of three (red, green and blue) individual component images. For this reason many of the techniques developed for monochrome images can be extended to color images by processing the three component images individually.

An image may be continuous with respect to the x - and y - coordinates and also in amplitude. Converting such an image to digital form requires that the coordinates, as well as the amplitude, be digitized.

APPLICATIONS OF DIGITAL IMAGE PROCESSING

Since digital image processing has very wide applications and almost all of the technical fields are impacted by DIP, we will just discuss some of the major applications of DIP.

Digital image processing has a broad spectrum of applications, such as

- Remote sensing via satellites and other spacecrafts
- Image transmission and storage for business applications
- Medical processing,
- RADAR (Radio Detection and Ranging)
- SONAR(Sound Navigation and Ranging) and
- Acoustic image processing (The study of underwater sound is known as underwater acoustics or hydro acoustics.)
- Robotics and automated inspection of industrial parts.

Images acquired by satellites are useful in tracking of

- Earth resources;
- Geographical mapping;
- Prediction of agricultural crops,
- Urban growth and weather monitoring
- Flood and fire control and many other environmental applications.

Space image applications include:

- Recognition and analysis of objects contained in images obtained from deep space-probe missions.
- Image transmission and storage applications occur in broadcast television
- Teleconferencing
- Transmission of facsimile images(Printed documents and graphics) for office automation

Communication over computer networks

- Closed-circuit television based security monitoring systems and
- In military communications.

Medical applications:

- Processing of chest X- rays
- Cineangiograms
- Projection images of transaxial tomography and
- Medical images that occur in radiology nuclear magnetic resonance(NMR)
- Ultrasonic scanning

IMAGE PROCESSING TOOLBOX (IPT) is a collection of functions that extend the capability of the MATLAB numeric computing environment. These functions, and the

expressiveness of the MATLAB language, make many image-processing operations easy to write in a compact, clear manner, thus providing a ideal software prototyping environment for the solution of image processing problem.

Components of Image processing System:

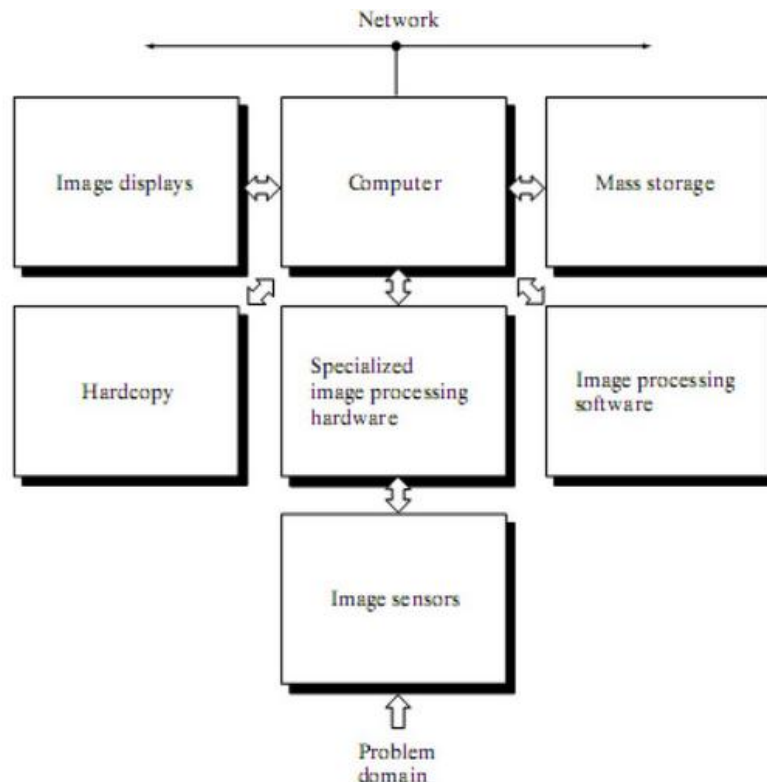


Figure : Components of Image processing System

Image Sensors: With reference to sensing, two elements are required to acquire digital image. The first is a physical device that is sensitive to the energy radiated by the object we wish to image and second is specialized image processing hardware.

Specialize image processing hardware: It consists of the digitizer just mentioned, plus hardware that performs other primitive operations such as an arithmetic logic unit, which performs arithmetic such addition and subtraction and logical operations in parallel on images.

Computer: It is a general purpose computer and can range from a PC to a supercomputer depending on the application. In dedicated applications, sometimes specially designed computer are used to achieve a required level of performance

Software: It consists of specialized modules that perform specific tasks a well designed package also includes capability for the user to write code, as a minimum, utilizes the specialized module. More sophisticated software packages allow the integration of these modules.

Mass storage: This capability is a must in image processing applications. An image of size 1024 x1024 pixels, in which the intensity of each pixel is an 8- bit quantity requires one Megabytes of storage space if the image is not compressed .Image processing applications falls into three principal categories of storage

- i) Short term storage for use during processing
- ii) On line storage for relatively fast retrieval
- iii) Archival storage such as magnetic tapes and disks

Image display: Image displays in use today are mainly color TV monitors. These monitors are driven by the outputs of image and graphics displays cards that are an integral part of computer system.

Hardcopy devices: The devices for recording image includes laser printers, film cameras, heat sensitive devices inkjet units and digital units such as optical and CD ROM disk. Films provide the highest possible resolution, but paper is the obvious medium of choice for written applications.

Networking: It is almost a default function in any computer system in use today because of the large amount of data inherent in image processing applications. The key consideration in image transmission bandwidth.

Fundamental Steps in Digital Image Processing:

There are two categories of the steps involved in the image processing –

- 1. Methods whose outputs are input are images.
- 2. Methods whose outputs are attributes extracted from those images.

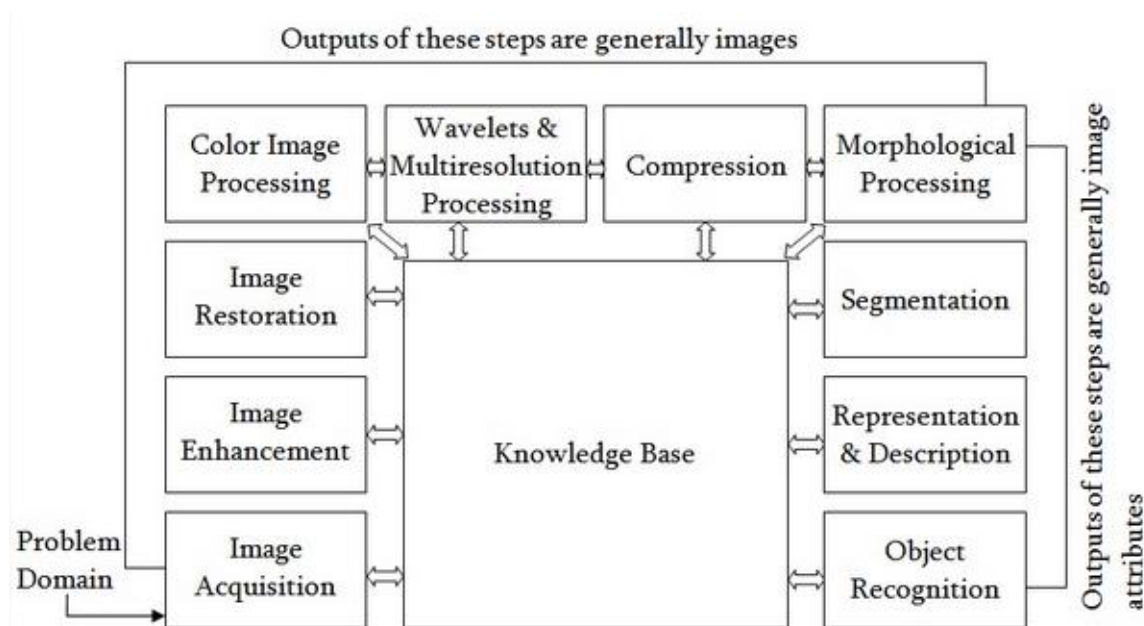


Fig: Fundamental Steps in Digital Image Processing

Image acquisition: It could be as simple as being given an image that is already in digital form. Generally the image acquisition stage involves processing such scaling.

Image Enhancement: It is among the simplest and most appealing areas of digital image processing. The idea behind this is to bring out details that are obscured or simply to highlight certain features of interest in image. Image enhancement is a very subjective area of image processing.

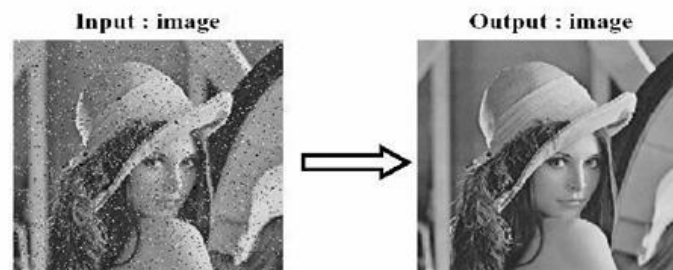
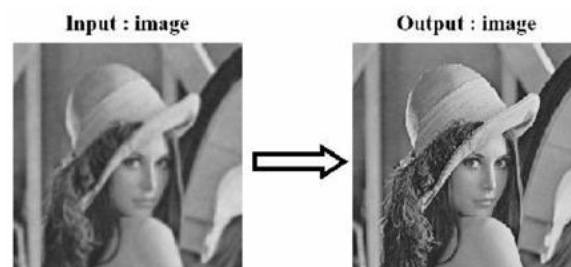


Image Restoration: It deals with improving the appearance of an image. It is an objective approach, in the sense that restoration techniques tend to be based on mathematical or probabilistic models of image processing. Enhancement, on the other hand is based on human subjective preferences regarding what constitutes a “good” enhancement result.



Color image processing: It is an area that is been gaining importance because of the use of digital images over the internet. Color image processing deals with basically color models and their implementation in image processing applications.

Wavelets and Multiresolution Processing: These are the foundation for representing image in various degrees of resolution.

Compression: It deals with techniques reducing the storage required to save an image, or the bandwidth required to transmit it over the network. It has two major approaches a) Lossless Compression b) Lossy Compression

Morphological processing: It deals with tools for extracting image components that are useful in the representation and description of shape and boundary of objects. It is majorly used in automated inspection applications.

Representation and Description: It always follows the output of segmentation step that is, raw pixel data, constituting either the boundary of an image or points in the region itself. In either case converting the data to a form suitable for computer processing is necessary.

Recognition: It is the process that assigns label to an object based on its descriptors. It is the last step of image processing which use artificial intelligence of software.

Knowledge base:

Knowledge about a problem domain is coded into an image processing system in the form of a knowledge base. This knowledge may be as simple as detailing regions of an image where the information of the interest is known to be located. Thus limiting search that has to be conducted in seeking the information. The knowledge base also can be quite complex such as an interrelated list of all major possible defects in a materials inspection problems or an image database containing high resolution satellite images of a region in connection with change detection application.

A Simple Image Model:

An image is denoted by a two dimensional function of the form $f\{x, y\}$. The value or amplitude of f at spatial coordinates $\{x,y\}$ is a positive scalar quantity whose physical meaning is determined by the source of the image. When an image is generated by a physical process, its values are proportional to energy radiated by a physical source. As a consequence, $f(x,y)$ must be nonzero and finite; that is $0 < f(x,y) < \infty$. The function $f(x,y)$ may be characterized by two components- The amount of the source illumination incident on the scene being viewed.

(a) The amount of the source illumination reflected back by the objects in the scene. These are called illumination and reflectance components and are denoted by $i(x,y)$ and $r(x,y)$ respectively.

The functions combine as a product to form $f(x,y)$. We call the intensity of a monochrome image at any coordinates (x,y) the gray level (I) of the image at that point $I = f(x, y)$

$$L_{\min} \leq I \leq L_{\max}$$

$L_{\min}$ is to be positive and $L_{\max}$ must be finite

$$L_{\min} = i_{\min} r_{\min}$$

$$L_{\max} = i_{\max} r_{\max}$$

The interval $[L_{\min}, L_{\max}]$ is called gray scale. Common practice is to shift this interval numerically to the interval $[0, L-1]$ where $I=0$ is considered black and $I= L-1$ is considered

white on the gray scale. All intermediate values are shades of gray of gray varying from black to white.

SAMPLING AND QUANTIZATION:

To create a digital image, we need to convert the continuous sensed data into digital form. This involves two processes – sampling and quantization. An image may be continuous with respect to the x and y coordinates and also in amplitude. To convert it into digital form we have to sample the function in both coordinates and in amplitudes.

Digitalizing the coordinate values is called sampling. Digitalizing the amplitude values is called quantization. There is a continuous the image along the line segment AB. To sample this function, we take equally spaced samples along line AB. The location of each sample is given by a vertical tick mark (mark) in the bottom part. The samples are shown as block squares superimposed on function the set of these discrete locations gives the sampled function.

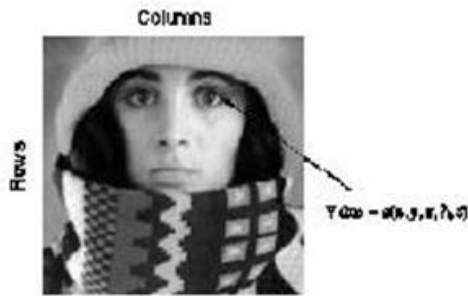
In order to form a digital, the gray level values must also be converted (quantized) into discrete quantities. So we divide the gray level scale into eight discrete levels ranging from eight level values. The continuous gray levels are quantized simply by assigning one of the eight discrete gray levels to each sample. The assignment is made depending on the vertical proximity of a sample to a vertical tick mark.

Starting at the top of the image and covering out this procedure line by line produces a two dimensional digital image.

Digital Image definition:

A digital image $f(m,n)$ described in a 2D discrete space is derived from an analog image $f(x,y)$ in a 2D continuous space through a sampling process that is frequently referred to as digitization. The mathematics of that sampling process will be described in subsequent Chapters. For now we will look at some basic definitions associated with the digital image. The effect of digitization is shown in figure.

The 2D continuous image $f(x,y)$ is divided into N rows and M columns. The intersection of a row and a column is termed a pixel. The value assigned to the integer coordinates (m,n) with $m=0,1,2..N-1$ and $n=0,1,2...N-1$ is $f(m,n)$. In fact, in most cases, is actually a function of many variables including depth, color and time (t).



There are three types of computerized processes in the processing of image

- 1) Low level process -these involve primitive operations such as image processing to reduce noise, contrast enhancement and image sharpening. These kind of processes are characterized by fact the both inputs and output are images.
- 2) Mid level image processing - it involves tasks like segmentation, description of those objects to reduce them to a form suitable for computer processing, and classification of individual objects. The inputs to the process are generally images but outputs are attributes extracted from images.
- 3) High level processing – It involves “making sense” of an ensemble of recognized objects, as in image analysis, and performing the cognitive functions normally associated with vision.

Representing Digital Images:

The result of sampling and quantization is matrix of real numbers. Assume that an image $f(x,y)$ is sampled so that the resulting digital image has M rows and N Columns. The values of the coordinates (x,y) now become discrete quantities thus the value of the coordinates at origin become $(0,0)$ The next Coordinates value along the first signify the iamge along the first row. it does not mean that these are the actual values of physical coordinates when the image was sampled.

$$f(x,y) \approx \begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,M-1) \\ f(1,0) & f(1,1) & \dots & f(1,M-1) \\ \vdots & \vdots & \ddots & \vdots \\ f(N-1,0) & f(N-1,1) & \dots & f(N-1,M-1) \end{bmatrix}$$

Thus the right side of the matrix represents a digital element, pixel or pel. The matrix can be represented in the following form as well. The sampling process may be viewed as partitioning the xy plane into a grid with the coordinates of the center of each grid being a pair of elements from the Cartesian products Z^2 which is the set of all ordered pair of elements (Z_i, Z_j) with Z_i and Z_j being integers from Z . Hence $f(x,y)$ is a digital image if gray

level (that is, a real number from the set of real number R) to each distinct pair of coordinates (x,y) . This functional assignment is the quantization process. If the gray levels are also integers, Z replaces R , the and a digital image become a 2D function whose coordinates and she amplitude value are integers. Due to processing storage and hardware consideration, the number gray levels typically is an integer power of 2.

$$L=2^k$$

Then, the number, b , of bites required to store a digital image is $B=M *N* k$ When $M=N$, the equation become $b=N^2*k$

When an image can have 2^k gray levels, it is referred to as “ k - bit”. An image with 256 possible gray levels is called an “8- bit image” ($256=2^8$).

Spatial and Gray level resolution:

Spatial resolution is the smallest discernible details are an image. Suppose a chart can be constructed with vertical lines of width w with the space between the also having width W , so a line pair consists of one such line and its adjacent space thus. The width of the line pair is $2w$ and there is $1/2w$ line pair per unit distance resolution is simply the smallest number of discernible line pair unit distance.

Gray levels resolution refers to smallest discernible change in gray levels. Measuring discernible change in gray levels is a highly subjective process reducing the number of bits R while repairing the spatial resolution constant creates the problem of false contouring.

It is caused by the use of an insufficient number of gray levels on the smooth areas of the digital image . It is called so because the rides resemble top graphics contours in a map. It is generally quite visible in image displayed using 16 or less uniformly spaced gray levels.

Image sensing and Acquisition:

The types of images in which we are interested are generated by the combination of an “illumination” source and the reflection or absorption of energy from that source by the elements of the “scene” being imaged. We enclose *illumination* and *scene* in quotes to emphasize the fact that they are considerably more general than the familiar situation in which a visible light source illuminates a common everyday 3-D (three-dimensional) scene. For example, the illumination may originate from a source of electromagnetic energy such as radar, infrared, or X-ray energy. But, as noted earlier, it could originate from less traditional sources, such as ultrasound or even a computer-generated illumination pattern. Similarly, the scene elements could be familiar objects, but they can just as easily be molecules, buried rock formations, or a human brain. We could even image a source, such as acquiring images

of the sun. Depending on the nature of the source, illumination energy is reflected from, or transmitted through, objects. An example in the first category is light reflected from a planar surface. An example in the second category is when X-rays pass through a patient's body for the purpose of generating a diagnostic X-ray film. In some applications, the reflected or transmitted energy is focused onto a photo converter (e.g., a phosphor screen), which converts the energy into visible light. Electron microscopy and some applications of gamma imaging use this approach. The idea is simple: Incoming energy is transformed into a voltage by the combination of input electrical power and sensor material that is responsive to the particular type of energy being detected. The output voltage waveform is the response of the sensor(s), and a digital quantity is obtained from each sensor by digitizing its response. In this section, we look at the principal modalities for image sensing and generation.

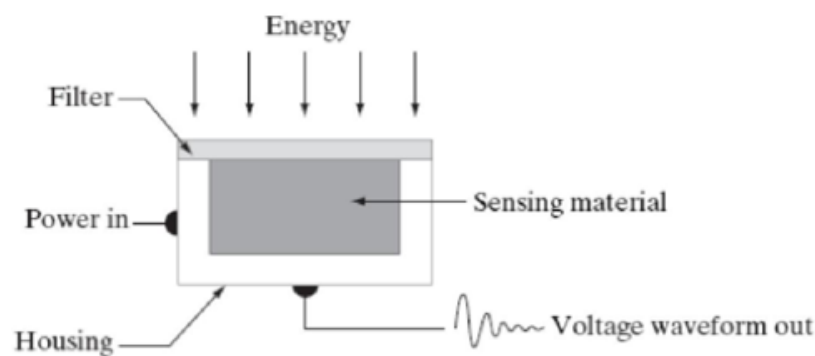


Fig:Single Image sensor



Fig: Line Sensor

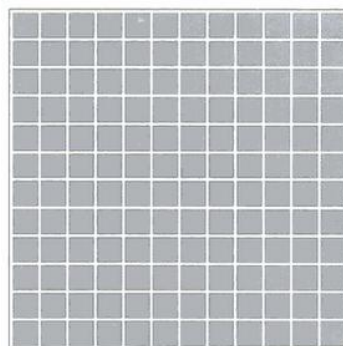
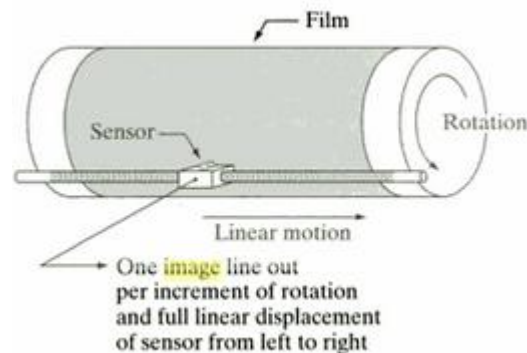


Fig: Array sensor

Image Acquisition using a Single sensor:

The components of a single sensor. Perhaps the most familiar sensor of this type is the photodiode, which is constructed of silicon materials and whose output voltage waveform is proportional to light. The use of a filter in front of a sensor improves selectivity. For

example, a green (pass) filter in front of a light sensor favors light in the green band of the color spectrum. As a consequence, the sensor output will be stronger for green light than for other components in the visible spectrum.



In order to generate a 2-D image using a single sensor, there has to be relative displacements in both the x- and y-directions between the sensor and the area to be imaged. Figure shows an arrangement used in high-precision scanning, where a film negative is mounted onto a drum whose mechanical rotation provides displacement in one dimension. The single sensor is mounted on a lead screw that provides motion in the perpendicular direction. Since mechanical motion can be controlled with high precision, this method is an inexpensive (but slow) way to obtain high-resolution images. Other similar mechanical arrangements use a flat bed, with the sensor moving in two linear directions. These types of mechanical digitizers sometimes are referred to as microdensitometers.

Image Acquisition using a Sensor strips:

A geometry that is used much more frequently than single sensors consists of an in-line arrangement of sensors in the form of a sensor strip, shows. The strip provides imaging elements in one direction. Motion perpendicular to the strip provides imaging in the other direction. This is the type of arrangement used in most flat bed scanners. Sensing devices with 4000 or more in-line sensors are possible. In-line sensors are used routinely in airborne imaging applications, in which the imaging system is mounted on an aircraft that flies at a constant altitude and speed over the geographical area to be imaged. One dimensional imaging sensor strips that respond to various bands of the electromagnetic spectrum are mounted perpendicular to the direction of flight. The imaging strip gives one line of an image at a time, and the motion of the strip completes the other dimension of a two-dimensional image. Lenses or other focusing schemes are used to project area to be scanned onto the sensors. Sensor strips mounted in a ring configuration are used in medical and industrial imaging to obtain cross-sectional (“slice”) images of 3-D objects.

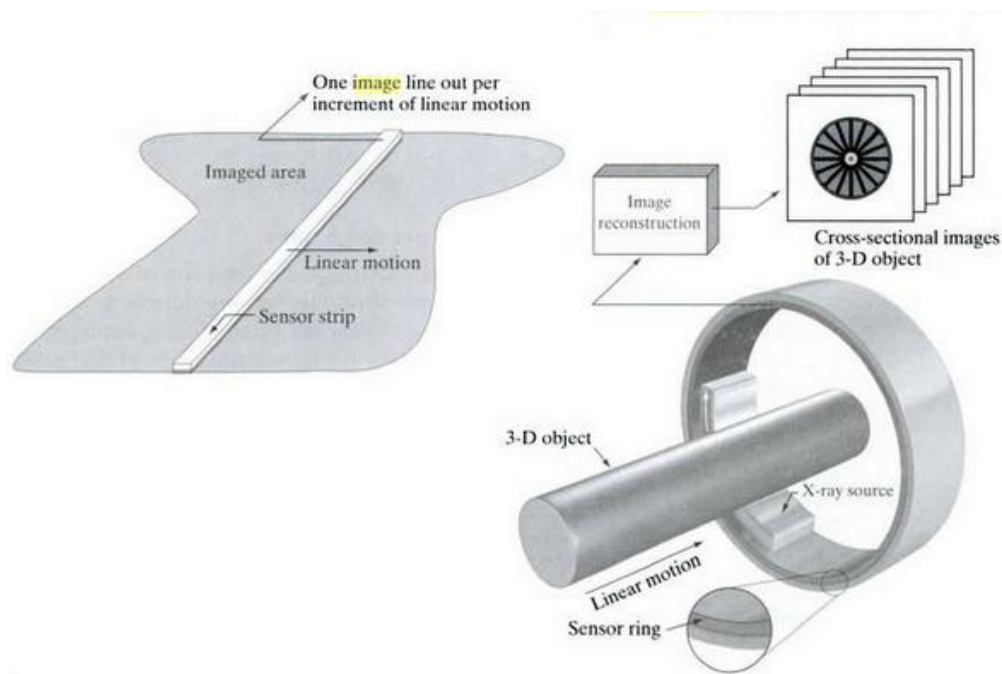


Fig: Image Acquisition using linear strip and circular strips.

Image Acquisition using a Sensor Arrays:

The individual sensors arranged in the form of a 2-D array. Numerous electromagnetic and some ultrasonic sensing devices frequently are arranged in an array format. This is also the predominant arrangement found in digital cameras. A typical sensor for these cameras is a CCD array, which can be manufactured with a broad range of sensing properties and can be packaged in rugged arrays of elements or more. CCD sensors are used widely in digital cameras and other light sensing instruments. The response of each sensor is proportional to the integral of the light energy projected onto the surface of the sensor, a property that is used in astronomical and other applications requiring low noise images. Noise reduction is achieved by letting the sensor integrate the input light signal over minutes or even hours. The two dimensional, its key advantage is that a complete image can be obtained by focusing the energy pattern onto the surface of the array. Motion obviously is not necessary, as is the case with the sensor arrangements This figure shows the energy from an illumination source being reflected from a scene element, but, as mentioned at the beginning of this section, the energy also could be transmitted through the scene elements. The first function performed by the imaging system is to collect the incoming energy and focus it onto an image plane. If the illumination is light, the front end of the imaging system is a lens, which projects the viewed scene onto the lens focal plane. The sensor array, which is coincident with the focal plane, produces outputs proportional to the integral of the light received at each sensor. Digital and

analog circuitry sweep these outputs and convert them to a video signal, which is then digitized by another section of the imaging system.

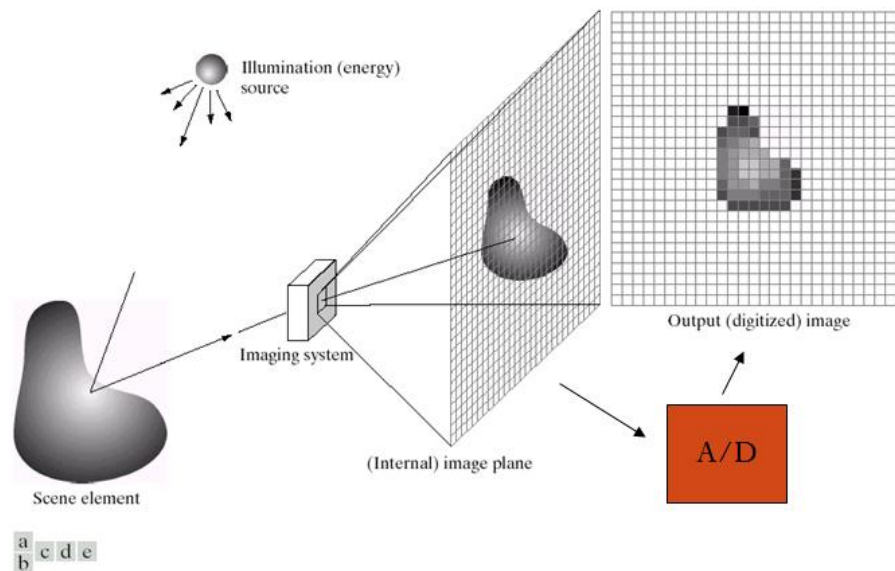
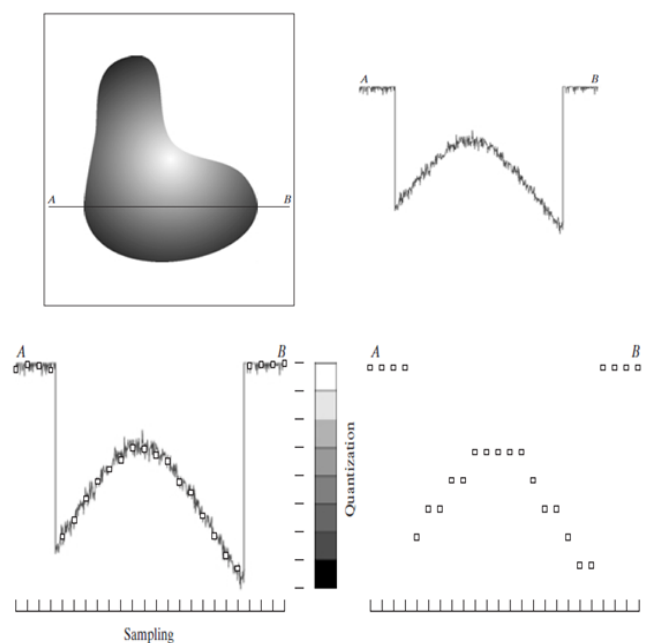


FIGURE An example of the digital image acquisition process. (a) Energy (“illumination”) source, (b) An element of a scene, (c) Imaging system, (d) Projection of the scene onto the image plane, (e) Digitized image.

Image sampling and Quantization:

To create a digital image, we need to convert the continuous sensed data into digital form. This involves two processes: *sampling* and *quantization*. A continuous image, $f(x, y)$, that we want to convert to digital form. An image may be continuous with respect to the x - and y -coordinates, and also in amplitude. To convert it to digital form, we have to sample the function in both coordinates and in amplitude. Digitizing the coordinate values is called *sampling*. Digitizing the amplitude values is called *quantization*.



Digital Image representation:

Digital image is a finite collection of discrete samples (*pixels*) of any observable object. The pixels represent a two- or higher dimensional “view” of the object, each pixel having its own discrete value in a finite range. The pixel values may represent the amount of visible light, infra red light, absorption of x-rays, electrons, or any other measurable value such as ultrasound wave impulses. The image does not need to have any visual sense; it is sufficient that the samples form a two-dimensional spatial structure that may be illustrated as an image. The images may be obtained by a digital camera, scanner, electron microscope, ultrasound stethoscope, or any other optical or non-optical sensor. Examples of digital image are:

- digital photographs
- satellite images
- radiological images (x-rays, mammograms)
- binary images, fax images, engineering drawings

Computer graphics, CAD drawings, and vector graphics in general are not considered in this course even though their reproduction is a possible source of an image. In fact, one goal of intermediate level image processing may be to reconstruct a model (e.g. vector representation) for a given digital image.

RELATIONSHIP BETWEEN PIXELS:

We consider several important relationships between pixels in a digital image.

NEIGHBORS OF A PIXEL

- A pixel p at coordinates (x,y) has four *horizontal* and *vertical* neighbors whose coordinates are given by:

$$(x+1,y), (x-1, y), (x, y+1), (x,y-1)$$

	$(x, y-1)$	
$(x-1, y)$	$P(x,y)$	$(x+1, y)$
	$(x, y+1)$	

This set of pixels, called the *4-neighbors* or p , is denoted by $N_4(p)$. Each pixel is one unit distance from (x,y) and some of the neighbors of p lie outside the digital image if (x,y) is on the border of the image. The four *diagonal* neighbors of p have coordinates and are denoted by $N_D(p)$.

$$(x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)$$

$(x-1, y+1)$		$(x+1, y-1)$
	$P(x, y)$	
$(x-1, y-1)$		$(x+1, y+1)$

These points, together with the 4-neighbors, are called the 8-neighbors of p , denoted by $N_8(p)$.

$(x-1, y+1)$	$(x, y-1)$	$(x+1, y-1)$
$(x-1, y)$	$P(x, y)$	$(x+1, y)$
$(x-1, y-1)$	$(x, y+1)$	$(x+1, y+1)$

As before, some of the points in $N_D(p)$ and $N_8(p)$ fall outside the image if (x, y) is on the border of the image.

ADJACENCY AND CONNECTIVITY

Let v be the set of gray-level values used to define adjacency, in a binary image, $v=\{1\}$.

In a gray-scale image, the idea is the same, but V typically contains more elements, for example, $V = \{180, 181, 182, \dots, 200\}$.

If the possible intensity values 0 – 255, V set can be any subset of these 256 values.

if we are reference to adjacency of pixel with value.

Three types of adjacency

- 4- Adjacency – two pixel P and Q with value from V are 4 –adjacency if A is in the set $N_4(P)$
- 8- Adjacency – two pixel P and Q with value from V are 8 –adjacency if A is in the set $N_8(P)$
- M-adjacency –two pixel P and Q with value from V are m – adjacency if (i) Q is in $N_4(p)$ or (ii) Q is in $N_D(q)$ and the set $N_4(p) \cap N_4(q)$ has no pixel whose values are from V .
- Mixed adjacency is a modification of 8-adjacency. It is introduced to eliminate the ambiguities that often arise when 8-adjacency is used.
- For example:

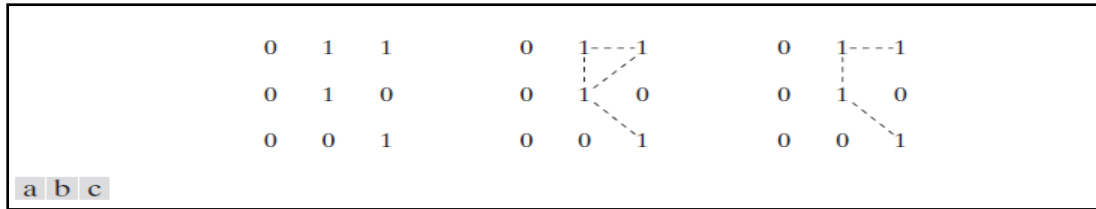


Fig:1.8(a) Arrangement of pixels; (b) pixels that are 8-adjacent (shown dashed) to the center pixel; (c) *m*-adjacency.

Types of Adjacency:

- In this example, we can note that to connect between two pixels (finding a path between two pixels):
 - In 8-adjacency way, you can find multiple paths between two pixels
 - While, in m-adjacency, you can find only one path between two pixels
- So, m-adjacency has eliminated the multiple path connection that has been generated by the 8-adjacency.
- Two subsets S_1 and S_2 are adjacent, if some pixel in S_1 is adjacent to some pixel in S_2 . Adjacent means, either 4-, 8- or m-adjacency.

A Digital Path:

- A digital path (or curve) from pixel p with coordinate (x,y) to pixel q with coordinate (s,t) is a sequence of distinct pixels with coordinates $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ where $(x_0, y_0) = (x, y)$ and $(x_n, y_n) = (s, t)$ and pixels (x_i, y_i) and (x_{i-1}, y_{i-1}) are adjacent for $1 \leq i \leq n$
- n is the length of the path
- If $(x_0, y_0) = (x_n, y_n)$, the path is closed.

We can specify 4-, 8- or m-paths depending on the type of adjacency specified.

- Return to the previous example:

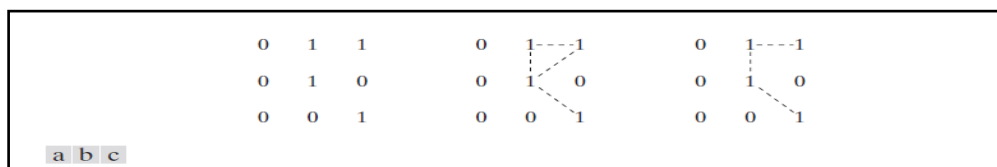


Fig:1.8 (a) Arrangement of pixels; (b) pixels that are 8-adjacent(shown dashed) to the center pixel; (c) m-adjacency.

In figure (b) the paths between the top right and bottom right pixels are 8-paths. And the path between the same 2 pixels in figure (c) is m-path

Connectivity:

- Let S represent a subset of pixels in an image, two pixels p and q are said to be connected in S if there exists a path between them consisting entirely of pixels in S .

- For any pixel p in S , the set of pixels that are connected to it in S is called a *connected component* of S . If it only has one connected component, then set S is called a *connected set*.

Region and Boundary:

- **REGION:** Let R be a subset of pixels in an image, we call R a region of the image if R is a connected set.
- **BOUNDARY:** The *boundary* (also called *border* or *contour*) of a region R is the set of pixels in the region that have one or more neighbors that are not in R .

If R happens to be an entire image, then its boundary is defined as the set of pixels in the first and last rows and columns in the image. This extra definition is required because an image has no neighbors beyond its borders. Normally, when we refer to a region, we are referring to subset of an image, and any pixels in the boundary of the region that happen to coincide with the border of the image are included implicitly as part of the region boundary.

DISTANCE MEASURES:

For pixel p, q and z with coordinate (x, y) , (s, t) and (v, w) respectively D is a distance function or metric if

$$D[p, q] \geq 0 \quad \{D[p, q] = 0 \text{ iff } p=q\}$$

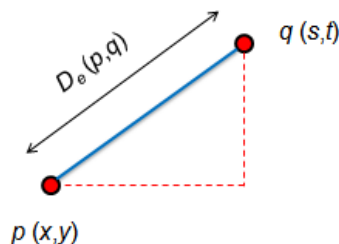
$$D[p, q] = D[q, p] \text{ and}$$

$$D[p, q] \geq 0 \quad \{D[p, q] + D(q, z)\}$$

- The **Euclidean Distance** between p and q is defined as:

$$D_e(p, q) = [(x - s)^2 + (y - t)^2]^{1/2}$$

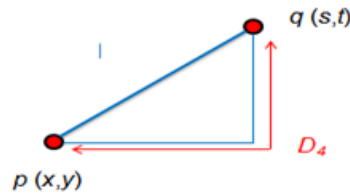
Pixels having a distance less than or equal to some value r from (x, y) are the points contained in a disk of radius r , centered at (x, y)



- The D_4 distance (also called **city-block distance**) between p and q is defined as:

$$D_4(p, q) = |x - s| + |y - t|$$

Pixels having a D_4 distance from (x,y) , less than or equal to some value r form a Diamond centered at (x,y)



Example:

The pixels with distance $D_4 \leq 2$ from (x,y) form the following contours of constant distance.

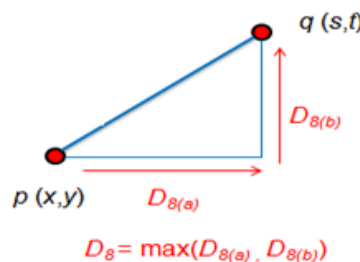
The pixels with $D_4 = 1$ are the 4-neighbors of (x,y)

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

- The D_8 distance (also called **chessboard distance**) between p and q is defined as:

$$D_8(p,q) = \max(|x-s|, |y-t|)$$

Pixels having a D_8 distance from (x,y) , less than or equal to some value r form a square Centered at (x,y) .



Example:

D_8 distance ≤ 2 from (x,y) form the following contours of constant distance.

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

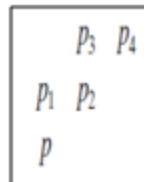
- D_m distance:**

It is defined as the shortest m -path between the points.

In this case, the distance between two pixels will depend on the values of the pixels along the path, as well as the values of their neighbors.

- Example:

Consider the following arrangement of pixels and assume that p , p_2 , and p_4 have value 1 and that p_1 and p_3 can have a value of 0 or 1. Suppose that we consider the adjacency of pixels values 1 (i.e. $V = \{1\}$)



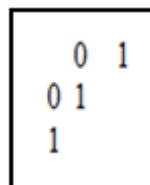
Now, to compute the D_m between points p and p_4

Here we have 4 cases:

Case1: If $p_1 = 0$ and $p_3 = 0$

The length of the shortest m-path

(the D_m distance) is 2 (p, p_2, p_4)

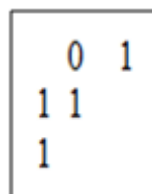


Case2: If $p_1 = 1$ and $p_3 = 0$

now, p_1 and p will no longer be adjacent (see m-adjacency definition)

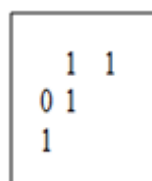
then, the length of the shortest

path will be 3 (p, p_1, p_2, p_4)



Case3: If $p_1 = 0$ and $p_3 = 1$

The same applies here, and the shortest m-path will be 3 (p, p_2, p_3, p_4)



Case4: If $p_1 = 1$ and $p_3 = 1$

The length of the shortest m-path will be 4 (p, p_1, p_2, p_3, p_4)

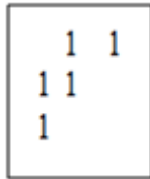


Image Storage & File Formats

File Format

- A file format is a standard way that information is encoded for storage in a computer file.
- A file format specifies how bits are used to encode information in a digital storage medium.
- File format may be either proprietary or free and may be either published or open.

Image File Format

It describes how data related to the image will be stored. Data can be stored in compressed, Uncompressed or vector format. Each format of the image have a different advantage and disadvantage.

Need for File Format

Sometimes image may be transmitted from one type of environment and may be accessed in different environment . It may cause interpretation mistakes therefore file format is required.

Types of File Format

- Simplest way of storing image data is by using 2D array of pixel intensities. This is known as bitmap.
- Another way is use of vector graphics where images are stored as collection of vectors. This format is suitable for synthetic images and line drawing.

Some of the raster file formats that are very popular are:

- GIF (Graphics Interchange Format)
- JPEG (Joint Photographic Experts Group)
- PNG (Portable Network Graphics)
- DICOM (Digital Imaging and Communication)

GIF (Graphics Interchange Format)

- It uses a lossless compression known as Lempel-Ziv-Welch (LZW) algorithm so the quality of images is very high.
- GIF files are used for web graphics. It also supports animation.
- They can be animated and are limited to only 256 colors, can allow for transparency. GIF files are typically small in size and are portable.

JPEG (Joint Photographic Expert Group)

- Another popular format for storing continuous tone images.
- It supports both lossy and lossless compression modes.
- It is very common format for storing and transmitting images on the World Wide Web.

PNG (Portable Network Graphics)

- It is a lossless image format.
- It was designed to replace gif format as gif supported 256 colors unlike PNG which support 16 million colors.

DICOM (Digital Imaging and Communication)

- It is popular file format in medical imaging.
- This format contains image data and metadata like patient details, equipments and acquisition details.

SVG (Scalable Vector Graphics)

- It is a vector graphics file format that enables 2D images to be displayed on web.
- These images are scalable to the size of viewing window and can be adjusted in size & resolution in which they are displayed.

Structure of File Format

Generally, file formats consist of two parts:

1. Image header

2. Image data

- Image header contains all required information to reconstruct a image using stored image data.
- It contains details about no. of pixels, version, data type, no. of bits used to represent pixel etc.
- Many types of formats are available such as BMP, GIF, PNG and JPEG. Some formats are open source so can be used frequently.

Structure of TIFF File Format

- The Tagged Image File Format (TIFF) is a standard format that is considered for the purpose of illustration.
- It stores image data without losing any data. It do not perform any compression on images, which is good for printing, professional printing.

The TIFF image format is as:

- **Image File Header (IFH)**
- **Image File Directory (IFD)**
- **Directory Entry (DE)**
- **Image Data**

TIFF Image Header

Bytes 0 & 1	Bytes 2 & 3	Bytes 4-7
• Byte Order	• Identification of File Format	• Offset to the First IFD

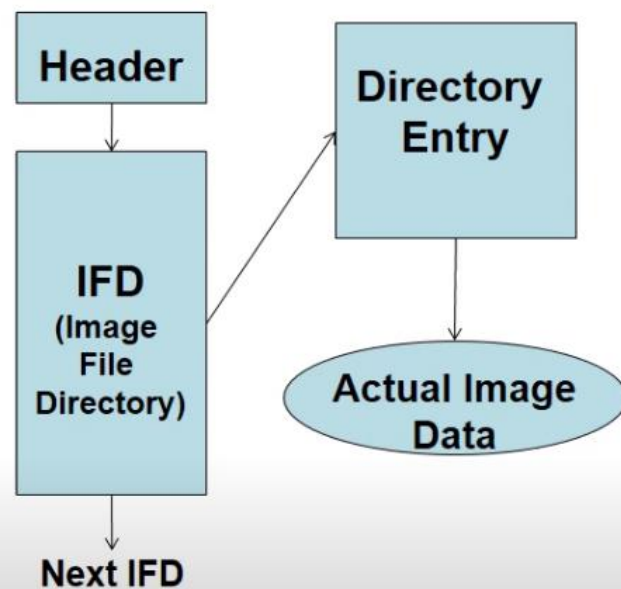
Image file directory (IFD) is a 12 bit entry that includes information on the counts of the DEs. If it is 0 it means there is no next IFD.

Every IFD is known as sub file and can store other related images.

Directory Entry (DE)

contains information about tags, counts and offset to the actual image data.

Once the image is acquired and stored in the system, many image processing operations can be applied.



TIFF Structure

- Need for transform
- 2D Orthogonal and Unitary transform and its properties
- 1D & 2D DFT – Properties – separability, translation, periodicity, conjugate symmetry, rotation, scaling, convolution and correlation
- Separable transforms
 - Walsh
 - Hadamard
 - Haar
 - Discrete Sine
 - DCT
 - Slant
 - SVD
 - KL transforms .

Need for transform

The need for transform is most of the signals or images are time domain signal (ie) signals can be measured with a function of time. This representation is not always best. For most image processing applications anyone of the mathematical transformation are applied to the signal or images to obtain further information from that signal.

2D Orthogonal and Unitary transform and its properties

As a one dimensional signal can be represented by an orthonormal set of basis vectors, an image can also be expanded in terms of a discrete set of basis arrays called basis images through a two dimensional (image) transform. For an $N \times N$ image $f(x,y)$ the forward and inverse transforms are given below

$$g(u,v) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} T(u,v,x,y) f(x,y)$$
$$f(x,y) = \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} I(x,y,u,v) g(u,v)$$

where, again, $T(u,v,x,y)$ and $I(x,y,u,v)$ are called the **forward and inverse transformation kernels**, respectively.

The forward kernel is said to be **separable** if

$$T(u,v,x,y) = T_1(u,x)T_2(v,y)$$

It is said to be **symmetric** if T_1 is functionally equal to T_2 such that

$$T(u,v,x,y) = T_1(u,x)T_1(v,y)$$

The same comments are valid for the inverse kernel.

If the kernel $T(u,v,x,y)$ of an image transform is separable and symmetric, then the transform $g(u,v) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} T(u,v,x,y) f(x,y) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} T_1(u,x)T_1(v,y) f(x,y)$ can be written in matrix form as follows

$$\underline{g} = \underline{T}_1 \cdot \underline{f} \cdot \underline{T}_1^T$$

where $\underline{f}$ is the original image of size $N \times N$, and $\underline{T}_1$ is an $N \times N$ transformation matrix with elements $t_{ij} = T_1(i,j)$. If, in addition, $\underline{T}_1$ is a unitary matrix then the transform is called **separable unitary** and the original image is recovered through the relationship

$$\underline{f} = \underline{T}_1^{*T} \cdot \underline{g} \cdot \underline{T}_1$$

Fundamental properties of unitary transforms

The property of energy preservation

In the unitary transformation

$$\underline{g} = \underline{T} \cdot \underline{f}$$

it is easily proven that

$$\|\underline{g}\|^2 = \|\underline{f}\|^2$$

Thus, a unitary transformation preserves the signal energy. This property is called energy preservation property. This means that every unitary transformation is simply a rotation of the vector $\underline{f}$ in the N - dimensional vector space.

The property of energy compaction

Most unitary transforms pack a large fraction of the energy of the image into relatively few of the transform coefficients. This means that relatively few of the transform coefficients have significant values and these are the coefficients that are close to the origin (small index coefficients).

1D & 2D DFT

Any function that periodically reports itself can be expressed as a sum of sines and cosines of different frequencies each multiplied by a different coefficient, this sum is called Fourier series. Even the functions which are non periodic but whose area under the curve is finite can also be represented in such form; this is now called Fourier transform. A function represented in either of these forms and can be completely reconstructed via an inverse process with no loss of information.

The discrete Fourier transform (DFT) of a sequence $\{u(n), n = 0, \dots, N - 1\}$ is defined as

$$v(k) = \sum_{n=0}^{N-1} u(n) W_N^{kn}, \quad k = 0, 1, \dots, N - 1$$

where

$$W_N \triangleq \exp\left\{\frac{-j2\pi}{N}\right\}$$

The inverse transform is given by

$$u(n) = \frac{1}{N} \sum_{k=0}^{N-1} v(k) W_N^{-kn}, \quad n = 0, 1, \dots, N - 1$$

The two-dimensional DFT of an $N \times N$ image $\{u(m, n)\}$ is a separable transform defined as

$$v(k, l) = \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} u(m, n) W_N^{km} W_N^{ln}, \quad 0 \leq k, l \leq N-1$$

and the inverse transform is

$$u(m, n) = \frac{1}{N^2} \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} v(k, l) W_N^{-km} W_N^{-ln}, \quad 0 \leq m, n \leq N-1$$

The two-dimensional unitary DFT pair is defined as

$$v(k, l) = \frac{1}{N} \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} u(m, n) W_N^{km} W_N^{ln}, \quad 0 \leq k, l \leq N-1$$

$$u(m, n) = \frac{1}{N} \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} v(k, l) W_N^{-km} W_N^{-ln}, \quad 0 \leq m, n \leq N-1$$

In matrix notation this becomes

$$\mathbf{V} = \mathbf{F} \mathbf{U} \mathbf{F}$$

The properties of the two-dimensional unitary DFT are quite similar to the one-dimensional case and are summarized next.

Symmetric, unitary.

$$\mathcal{F}^T = \mathcal{F}, \quad \mathcal{F}^{-1} = \mathcal{F}^* = \mathbf{F}^* \otimes \mathbf{F}^*$$

Periodic extensions.

$$v(k + N, l + N) = v(k, l), \quad \forall k, l$$

$$u(m + N, n + N) = u(m, n), \quad \forall m, n$$

Sampled Fourier spectrum. If $\tilde{u}(m, n) = u(m, n)$, $0 \leq m, n \leq N-1$, and $\tilde{u}(m, n) = 0$ otherwise, then

$$\tilde{U}\left(\frac{2\pi k}{N}, \frac{2\pi l}{N}\right) = \text{DFT}\{u(m, n)\} = v(k, l)$$

where $\tilde{U}(\omega_1, \omega_2)$ is the Fourier transform of $\tilde{u}(m, n)$.

Fast transform. Since the two-dimensional DFT is separable, the transformation of $f(x, y)$ is equivalent to $2N$ one-dimensional unitary DFTs, each of which can be performed in $O(N \log_2 N)$ operations via the FFT. Hence the total number of operations is $O(N^2 \log_2 N)$.

Conjugate symmetry. The DFT and unitary DFT of *real images* exhibit conjugate symmetry, that is,

$$v\left(\frac{N}{2} \pm k, \frac{N}{2} \pm l\right) = v^*\left(\frac{N}{2} \mp k, \frac{N}{2} \mp l\right), \quad 0 \leq k, l \leq \frac{N}{2} - 1$$

THE DISCRETE COSINE TRANSFORM (DCT)

This is a transform that is similar to the Fourier transform in the sense that the new independent variable represents again frequency. The DCT is defined below.

$$C(u) = a(u) \sum_{x=0}^{N-1} f(x) \cos\left[\frac{(2x+1)u\pi}{2N}\right], \quad u = 0, 1, \dots, N-1$$

with $a(u)$ a parameter that is defined below.

$$a(u) = \begin{cases} \sqrt{1/N} & u = 0 \\ \sqrt{2/N} & u = 1, \dots, N-1 \end{cases}$$

The inverse DCT (IDCT) is defined below.

$$f(x) = \sum_{u=0}^{N-1} a(u) C(u) \cos\left[\frac{(2x+1)u\pi}{2N}\right]$$

Two dimensional signals (images)

$$C(u, v) = a(u)a(v) \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \cos\left[\frac{(2x+1)u\pi}{2N}\right] \cos\left[\frac{(2y+1)v\pi}{2N}\right]$$

$$f(x, y) = \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} a(u)a(v) C(u, v) \cos\left[\frac{(2x+1)u\pi}{2N}\right] \cos\left[\frac{(2y+1)v\pi}{2N}\right]$$

Properties of the DCT transform

- The cosine transform is real and orthogonal.
- The cosine transform is not a real part of the unitary DFT.
- The cosine transform of a sequence is related to the DFT of its antisymmetric extension
- The cosine transform is a fast transform
- The basis vectors of the cosine transform are the eigen vectors of the symmetric tridiagonal of Toeplitz matrix

- The cosine transform is close to the KL transform of first order Markov sequences. The cosine transform has very good to excellent energy compaction property of images,

The DCT is a real transform. This property makes it attractive in comparison to the Fourier transform.

The DCT has excellent energy compaction properties. For that reason it is widely used in image compression standards (as for example JPEG standards).

There are fast algorithms to compute the DCT, similar to the FFT for computing the DFT.

Sine Transform

The $N \times N$ sine transform matrix $\Psi = \{\psi(k, n)\}$, also called the *discrete sine transform* (DST), is defined as

$$\psi(k, n) = \sqrt{\frac{2}{N+1}} \sin \frac{\pi(k+1)(n+1)}{N+1}, \quad 0 \leq k, n \leq N-1$$

The sine transform pair of one-dimensional sequences is defined as

$$v(k) = \sqrt{\frac{2}{N+1}} \sum_{n=0}^{N-1} u(n) \sin \frac{\pi(k+1)(n+1)}{N+1}, \quad 0 \leq k \leq N-1$$

$$u(n) = \sqrt{\frac{2}{N+1}} \sum_{k=0}^{N-1} v(k) \sin \frac{\pi(k+1)(n+1)}{N+1}, \quad 0 \leq n \leq N-1$$

Properties of the Discrete Sine Transform

- The sine transform is real, symmetric and orthogonal.
- The sine transform is not the imaginary part of the unitary DFT.
- The sine transform of a sequence is related to the DFT of its antisymmetric extension
- The sine transform is a fast transform
- The basis vectors of the sine transform are the eigen vectors of the symmetric tridiagonal of Toeplitz matrix
- The sine transform is close to the KL transform of first order Markov sequences. The sine transform has very good to excellent energy compaction property of images,

Hadamard Transform

In a similar form as the Walsh transform, the 2-D Hadamard transform is defined as follows.

Forward

$$H(u, v) = \frac{1}{N} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \left[\prod_{i=0}^{n-1} (-1)^{(b_i(x)b_i(u) + b_i(y)b_i(v))} \right], \quad N = 2^n \text{ or}$$
$$H(u, v) = \frac{1}{N} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) (-1)^{\sum_{i=0}^{n-1} (b_i(x)b_i(u) + b_i(y)b_i(v))}$$

Inverse

$$f(x, y) = \frac{1}{N} \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} H(u, v) \left[\prod_{i=0}^{n-1} (-1)^{(b_i(x)b_i(u) + b_i(y)b_i(v))} \right] \text{ etc.}$$

Properties of the Hadamard Transform

- The Hadamard transform is real, symmetric and orthogonal.
- The Hadamard transform is a fast transform
- The Hadamard transform has very good to excellent energy compaction property of images,

Properties are almost similar to that of *Walsh transform*.

The Hadamard transform differs from the Walsh transform only in the order of basis functions. The order of basis functions of the Hadamard transform **does not** allow the fast computation of it by using a straightforward modification of the FFT. An extended version of the Hadamard transform is the **Ordered Hadamard Transform** for which a fast algorithm called **Fast Hadamard Transform (FHT)** can be applied.

Walsh Transform

The discrete Walsh transform of a function $f(x)$, denoted $W(u)$, is given by

$$W(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) \prod_{j=0}^{n-1} (-1)^{b_j(x)b_{n-1-j}(u)}$$

Walsh transform kernel is symmetric matrix having orthogonal rows and columns. These properties, which hold in general, lead to an inverse kernel given by

$$h(x, u) = \prod_{j=0}^{n-1} (-1)^{b_j(x)b_{n-1-j}(u)}.$$

Thus the inverse Walsh transform is given by

$$f(x) = \sum_{u=0}^{N-1} W(u) \prod_{j=0}^{n-1} (-1)^{b_j(x)b_{n-1-j}(u)}.$$

The 2-D forward and inverse Walsh kernels are given by

$$g(x, y, u, v) = \frac{1}{N} \prod_{j=0}^{n-1} (-1)^{[b_j(x)b_{n-1-j}(u) + b_j(y)b_{n-1-j}(v)]}$$

and

$$h(x, y, u, v) = \frac{1}{N} \prod_{j=0}^{n-1} (-1)^{[b_j(x)b_{n-1-j}(u) + b_j(y)b_{n-1-j}(v)]}.$$

Thus the forward and inverse Walsh transforms for 2-D are given by

$$W(u, v) = \frac{1}{N} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \prod_{j=0}^{n-1} (-1)^{[b_j(x)b_{n-1-j}(u) + b_j(y)b_{n-1-j}(v)]}$$

and

$$f(x, y) = \frac{1}{N} \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} W(u, v) \prod_{i=0}^{n-1} (-1)^{[b_i(x)b_{n-1-i}(u) + b_i(y)b_{n-1-i}(v)]}$$

The Walsh Transform kernels are seperable and symmetric, because

$$\begin{aligned} g(x, y, u, v) &= g_1(x, u)g_1(y, v) \\ &= h_1(x, u)h_1(y, v) \\ &= \left[\frac{1}{\sqrt{N}} \prod_{i=0}^{n-1} (-1)^{b_i(x)b_{n-1-i}(u)} \right] \left[\frac{1}{\sqrt{N}} \prod_{i=0}^{n-1} (-1)^{b_i(y)b_{n-1-i}(v)} \right]. \end{aligned}$$

Values of the 1-D walsch transform kernel for N = 8 is

$\begin{array}{c} x \\ u \end{array}$	0	1	2	3	4	5	6	7
0	+	+	+	+	+	+	+	+
1	+	+	+	+	-	-	-	-
2	+	+	-	-	+	+	-	-
3	+	+	-	-	-	-	+	+
4	+	-	+	-	+	-	+	-
5	+	-	+	-	-	+	-	+
6	+	-	-	+	+	-	-	+
7	+	-	-	+	-	+	+	-

Properties of the Walsh Transform

- The Walsh transform is real, symmetric and orthogonal.
- The Walsh transform is a fast transform
- The Walsh transform has very good to excellent energy compaction property of images,
- □ Unlike the Fourier transform, which is based on trigonometric terms, the Walsh transform consists of a series expansion of basis functions whose values are only 0 or 1 and they have the form of square waves. These functions can be implemented more efficiently in a digital environment than the exponential basis functions of the Fourier transform.
- □ The forward and inverse Walsh kernels are identical except for a constant multiplicative factor of $\sqrt{N}$ for 1-D signals.
- □ The forward and inverse Walsh kernels are identical for 2-D signals. This is because the array formed by the kernels is a symmetric matrix having orthogonal rows and columns, so its inverse array is the same as the array itself.
- □ The concept of frequency exists also in Walsh transform basis functions. We can think of frequency as the number of zero crossings or the number of transitions in a basis vector and we call this number **sequency**. The Walsh transform exhibits the property of energy compaction as all the transforms that we are currently studying
- □ For the fast computation of the Walsh transform there exists an algorithm called **Fast Walsh Transform (FWT)**. This is a straightforward modification of the FFT.

Karhunen-Loeve Transform or KLT

The Karhunen-Loeve Transform or KLT was originally introduced as a series expansion for continuous random processes by Karhunen and Loeve. For discrete signals Hotelling first studied what was called a method of principal components, which is the discrete equivalent of the KL series expansion. Consequently, the KL transform is also called the Hotelling transform or the method of principal components.

Let $\underline{e}_i$ and $\lambda_i, i = 1, 2, \dots, n$, be this set of eigenvectors and corresponding eigenvalues of $\underline{C}_x$, arranged in descending order so that $\lambda_i \geq \lambda_{i+1}$ for $i = 1, 2, \dots, n-1$. Let $\underline{A}$ be a matrix whose rows are formed from the eigenvectors of $\underline{C}_x$, ordered so that the first row of $\underline{A}$ is the eigenvector corresponding to the largest eigenvalue, and the last row the eigenvector corresponding to the smallest eigenvalue.

Suppose that $\underline{A}$ is a transformation matrix that maps the vectors $\underline{x}$'s into vectors $\underline{y}$'s by using the following transformation

$$\underline{y} = \underline{A}(\underline{x} - \underline{m}_x)$$

The above transform is called the **Karhunen-Loeve** or **Hotelling** transform. The mean of the $\underline{y}$ vectors resulting from the above transformation is zero (try to prove that)

$$\underline{m}_y = \underline{0}$$

the covariance matrix is (try to prove that)

$$\underline{C}_y = \underline{A} \underline{C}_x \underline{A}^T$$

and $\underline{C}_y$ is a diagonal matrix whose elements along the main diagonal are the eigenvalues of $\underline{C}_x$ (try to prove that)

$$\underline{C}_y = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}$$

The off-diagonal elements of the covariance matrix are 0, so the elements of the $\underline{y}$ vectors are uncorrelated.

Lets try to reconstruct any of the original vectors $\underline{x}$ from its corresponding $\underline{y}$. Because the rows of $\underline{A}$ are orthonormal vectors (why?), then $\underline{A}^{-1} = \underline{A}^T$, and any vector $\underline{x}$ can be recovered from its corresponding vector $\underline{y}$ by using the relation

Properties of the Karhunen-Loeve transform

Despite its favourable theoretical properties, the KLT is not used in practice for the following reasons.

- Its basis functions depend on the covariance matrix of the image, and hence they have to be recomputed and transmitted for every image.
- Perfect decorrelation is not possible, since images can rarely be modelled as realisations of ergodic fields.
- There are no fast computational algorithms for its implementation.

Harr Transform

The harr function $h_k(x)$ are defined on a continuous interval, $x \in [0, 1]$, and for $k = 0$ to $N-1$, where $N=2^n$, The integer k can be uniquely decomposed as

$$K = 2^p + q - 1$$

Where $0 \leq p \leq n-1$

$q=0,1$ for $p=0$ and $1 \leq q \leq 2^p$.

For example, $N=4$ we have

k	0	1	2	3
p	0	0	1	1
q	0	1	1	2

The harr function can be defined as

$$h_0(x) \triangleq h_{0,0}(x) = \frac{1}{\sqrt{N}}, \quad x \in [0, 1].$$

$$h_k(x) \triangleq h_{p,q}(x) = \frac{1}{\sqrt{N}} \begin{cases} 2^{p/2}, & \frac{q-1}{2^p} \leq x < \frac{q-\frac{1}{2}}{2^p} \\ -2^{p/2}, & \frac{q-\frac{1}{2}}{2^p} \leq x < \frac{q}{2^p} \\ 0, & \text{otherwise for } x \in [0, 1] \end{cases}$$

For $N=8$. The harr transform is given by

$$\mathbf{Hr} = \frac{1}{\sqrt{8}} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{2} & \sqrt{2} & -\sqrt{2} & -\sqrt{2} \\ 2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & -2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 \end{bmatrix} \begin{matrix} 0 \\ 1 \\ 2 \\ 2 \\ 2 \\ 2 \\ 2 \\ 2 \end{matrix}$$

Sequency

Properties of Harr transform

- The Harr transform is real and orthogonal.
- The Harr transform is a fast transform
- The basis vectors of the Harr transform are sequency ordered
- The Harr transform is close to the KL transform of first order Markov sequences. The Harr transform has poor energy compaction property for images

Slant transform

The $N \times N$ slant transform matrices are defined by the recursion

$$S_n = \frac{1}{\sqrt{2}}$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ a_n & b_n & 0 & -a_n & b_n & 0 \\ 0 & I_{(N/2)-2} & 0 & I_{(N/2)-2} & S_{n-1} & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ -b_n & a_n & 0 & b_n & a_n & 0 \\ 0 & I_{(N/2)-2} & 0 & -I_{(N/2)-2} & 0 & S_{n-1} \end{bmatrix}$$

The 4x4 slant transformation matrix is obtained as

$$S_2 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ \frac{3}{\sqrt{5}} & \frac{1}{\sqrt{5}} & \frac{-1}{\sqrt{5}} & \frac{-3}{\sqrt{5}} \\ 1 & -1 & -1 & 1 \\ \frac{1}{\sqrt{5}} & \frac{-3}{\sqrt{5}} & \frac{3}{\sqrt{5}} & \frac{-1}{\sqrt{5}} \end{bmatrix} \quad \begin{array}{c} \text{Sequency} \\ 0 \\ 1 \\ 2 \\ 3 \end{array}$$

Properties of slant transform

- The slant transform is real and orthogonal.
- The slant transform is a fast transform
- The basis vectors of the slant transform are not sequency ordered for $n \geq 3$.
- The slant transform is close to the KL transform of first order Markov sequences. The slant transform is very good to excellent energy compaction property for images